PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	The Association between cytokine levels, sepsis severity, and clinical outcomes in sepsis: a quantitative systematic review protocol.
AUTHORS	Gharamti, Amal; Samara, Omar; Monzon, Anthony; Scherger, Sias; DeSanto, Kristen; Sillau, Stefan; Franco-Paredes, Carlos; Henao-Martínez, Andrés; Shapiro, Leland

VERSION 1 – REVIEW

REVIEWER	Besen, Bruno Universidade de Sao Paulo Faculdade de Medicina, Medical Emergencies Discipline
REVIEW RETURNED	13-Mar-2021

GENERAL COMMENTS	The authors present a protocol for a systematic review about cytokine levels and clinical outcomes in sepsis and septic shock. The systematic review protocol has been registered in PROSPERO under CRD42020179800. There is not much novelty in this research question, although a systematic review would be interesting to summarise current evidence. Regarding their research question, according to the PROGRESS framework, the authors seem to aim for conducting level IV prognostic research (i.e., predictors of treatment effect). I state this because their explanation for the research comes from the inability of cytokine blocking treatments to demonstrate better clinical outcomes in sepsis. However, by stating that they want to include observational studies, this makes the research question a bit confusing. Do they only want to study cytokine levels as a prognostic factor over and above clinically observable variables (such as organ dysfunction variables and routinely collected variables such as lymphocytes, leucocytes, C reactive protein, etc...); or do they want to evaluate whether cytokine levels are predictors of treatment effect in studies of cytokine blocking drugs? In my view, the authors should be more clear about their objectives regarding this systematic review. Stemming from the need to better delineate their research question (or at least put it on paper), the methods also have issues: 1) Study level variables can be easily used to explore heterogeneity in meta-analysis, but the authors pre-specify a number of individual-level covariates (whose mean value or proportion could be evaluated as a study level variable, although this is much harder to do). Do the authors want to explore study-level heterogeneity or individual level covariates? If it is the latter,

	then an individual patient data meta-analysis would be more recommended. 2) The outcomes presented in this protocol are somewhat different from the PROSPERO registry. For example, in the protocol you state 28-30 day mortality. In the registry, you state 28 day, 90-day and 1-year mortality. The types of studies to be included are also different from the PROSPERO registry. 3) Why will the authors use the 1992 Sepsis definitions instead of current definitions? Please explain. 4) The authors sometimes mix inclusion/exclusion criteria of individual level data instead of study-level data. Inclusion/exclusion criteria must refer to study characteristics, no individual level characteristics. Please revise (Ex: pages 179-181) 5) Since this is prognostic factor research, the manuscripts should be assessed for risk of bias using the QUIPS tool or the PROBAST tool, not Cochrane's RCT RoB tool neither the Newcastle-Ottawa Scale (NOS). Consider reading the Chapter 9 of the book "Prognosis Research in Healthcare" from Richard Riley and collaborators for further explanation. 6) In the data synthesis session, the authors also mix individual level covariates with study level aggregate measures (age, APACHE, etc...) 7) Finally, I suggest to the authors some piloting of their data extraction tools and meta-analysis to assess the real feasibility of such research question. Issues of protocol readability: 1) I would suggest to the authors to shorten their Introduction to be very straight to the point. In the protocol, the introduction is three-pages long and takes some time to get to the point. Thank you for the opportunity to review your manuscript.

REVIEWER	Conti, Pio University of Chieti-Pescara Postgraduate Medical School
REVIEW RETURNED	20-Mar-2021

GENERAL COMMENTS	Title: The Association between cytokine levels, sepsis severity, and clinical outcomes in sepsis: a 2 quantitative systematic review protocol. In this paper, the authors study the pro-inflammatory cytokines TNF, IL-1 and IFN in sepsis and evaluate the association between cytokine levels and sepsis-related morbidity and mortality. The abstract lacks a conclusion. Please enter. TNF should be written without alpha (with old fashion alpha). Throughout the manuscript Abstract, first sentence: I believe the opposite, which is that sepsis causes the release of pro-inflammatory cytokines and not that cytokines cause sepsis. To correct. When discussing pro-inflammatory cytokines such as IL-1 in sepsis, in the discussion, the authors should also mention the anti-inflammatory cytokines that are arousing great interest today. Therefore, to make this paper more interesting for the readers of

	this important journal, the authors should expand a little the discussion on this subject, in order to give a wider view to the reader. Below I list 3 interesting articles that should be studied, incorporate the meaning and report them briefly in the discussion, and in the list of references. Coronavirus-19 (SARS-CoV-2) induces acute severe lung inflammation via IL-1 causing cytokine storm in COVID-19: a promising inhibitory strategy. Conti P, Caraffa A, Gallenga CE, Ross R, Kritas SK, Frydas I, Younes A, Ronconi G.J Biol Regul Homeost Agents. 2020 Oct 5;34(6). Induction of pro-inflammatory cytokines (IL-1 and IL-6) and lung inflammation by Coronavirus-19 (COVI-19 or SARS-CoV-2): anti-inflammatory strategies. Conti P, Ronconi G, Caraffa A, Gallenga CE, Ross R, Frydas I, Kritas SK. J Biol Regul Homeost Agents. 2020 March-April,;34(2):327-331. Mast cells contribute to coronavirus-induced inflammation: new anti-inflammatory strategy. Kritas SK, Ronconi G, Caraffa A, et al. J Biol Regul Homeost Agents. 2020 January-February,;34(1):9-14. I believe these suggestions are important for improving this paper. Without these corrections the paper cannot be published. So I recommend minor revision. I'd like to review this article after corrections.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1:
Comments to the Author:
The authors present a protocol for a systematic review about cytokine levels and clinical outcomes in sepsis and septic shock. The systematic review protocol has been registered in PROSPERO under CRD42020179800. There is not much novelty in this research question, although a systematic review would be interesting to summarise current evidence.

- Regarding their research question, according to the PROGRESS framework, the authors seem to aim for conducting level IV prognostic research (i.e., predictors of treatment effect). I state this because their explanation for the research comes from the inability of cytokine blocking treatments to demonstrate better clinical outcomes in sepsis. However, by stating that they want to include observational studies, this makes the research question a bit confusing. Do they only want to study cytokine levels as a prognostic factor over and above clinically observable variables (such as organ dysfunction variables and routinely collected variables such as lymphocytes, leucocytes, C reactive protein, etc...); or do they want to evaluate whether cytokine levels are predictors of treatment effect in studies of cytokine blocking drugs? In my view, the authors should be more clear about their objectives regarding this systematic review.
  - I appreciate the reviewer for the feedback. We edited our Introduction and aimed to be more explicit about our objectives
  - We also abbreviated the Introduction and delete some of the statements regarding treatment effect in studies of cytokine blocking drugs to avoid confusion

Stemming from the need to better delineate their research question (or at least put it on paper), the methods also have issues:

1) Study level variables can be easily used to explore heterogeneity in meta-analysis, but the authors pre-specify a number of individual-level covariates (whose mean value or proportion could be evaluated as a study level variable, although this is much harder to do). Do the authors want to explore study-level heterogeneity or individual level covariates? If it is the latter, then an individual patient data meta-analysis would be more recommended.

- I appreciate the comment from the reviewer. Page 8, line 253: We amended the method section to specify that we will perform a study-level meta-analysis and explore the study-level heterogeneity.

2) The outcomes presented in this protocol are somewhat different from the PROSPERO registry. For example, in the protocol you state 28-30 day mortality. In the registry, you state 28 day, 90-day and 1-year mortality. The types of studies to be included are also different from the PROSPERO registry.

- The PROSPERO registry has been updated to match the current protocol.

3) Why will the authors use the 1992 Sepsis definitions instead of current definitions? Please explain.

- We used the 1992 Sepsis definitions because most studies that studied cytokine levels or cytokine-blocking therapies used the 1992 definition. If we used current definitions, we would limit extensively the number of manuscripts included. This clarification has been added to the manuscript.

4) The authors sometimes mix inclusion/exclusion criteria of individual level data instead of study-level data. Inclusion/exclusion criteria must refer to study characteristics, no individual level characteristics. Please revise (Ex: pages 179-181)

- We thank the reviewer for this comment, and inclusion/exclusion criteria have been amended to refer to study characteristics (page 8, line 258).

5) Since this is prognostic factor research, the manuscripts should be assessed for risk of bias using the QUIPS tool or the PROBAST tool, not Cochrane's RCT RoB tool neither the Newcastle-Ottawa Scale (NOS). Consider reading the Chapter 9 of the book "Prognosis Research in Healthcare" from Richard Riley and collaborators for further explanation.

- We thank the reviewer for this comment. The QUIPS tool will be used instead (page 7, line 237).

6) In the data synthesis session, the authors also mix individual level covariates with study level aggregate measures (age, APACHE, etc...)

- We thank the reviewer for this comment. We edited the variables to reflect the study level aggregate measures (page 8, line 258).

7) Finally, I suggest to the authors some piloting of their data extraction tools and meta-analysis to assess the real feasibility of such research question.

- We thank the reviewer for this comment. We edited our data extraction section to include piloting the data extraction tools and meta-analysis (page 7, line 228)

Issues of protocol readability:

1) I would suggest to the authors to shorten their Introduction to be very straight to the point. In the protocol, the Introduction is three-pages long and takes some time to get to the point.

- We have shortened our introduction section as recommended.

Reviewer 2:
Comments to the Author:
Title: The Association between cytokine levels, sepsis severity, and clinical outcomes in sepsis: a 2 quantitative systematic review protocol.

In this paper, the authors study the pro-inflammatory cytokines TNF, IL-1 and IFN in sepsis and evaluate the association between cytokine levels and sepsis-related morbidity and mortality.

The abstract lacks a conclusion. Please enter.
Comment from the Editor: Please note that Protocol articles should not include conclusion sections.

- Noted.

1) TNF should be written without alpha (with old fashion alpha). Throughout the manuscript.

- This has been changed throughout the manuscript.

2) Abstract, first sentence: I believe the opposite, which is that sepsis causes the release of pro-inflammatory cytokines and not that cytokines cause sepsis. To correct.

- The sentence was clarified as suggested

3) When discussing pro-inflammatory cytokines such as IL-1 in sepsis, in the discussion, the authors should also mention the anti-inflammatory cytokines that are arousing great interest today. Therefore, to make this paper more interesting for the readers of this important journal, the authors should expand a little the discussion on this subject, in order to give a wider view to the reader.

- We added a paragraph discussing the anti-inflammatory cytokines.

4) Below I list 3 interesting articles that should be studied, incorporate the meaning and report them briefly in the discussion, and in the list of references.

Coronavirus-19 (SARS-CoV-2) induces acute severe lung inflammation via IL-1 causing cytokine storm in COVID-19: a promising inhibitory strategy.Conti P, Caraffa A, Gallenga CE, Ross R, Kritas SK, Frydas I, Younes A, Ronconi G.J Biol Regul Homeost Agents. 2020 Oct 5;34(6).

Induction of pro-inflammatory cytokines (IL-1 and IL-6) and lung inflammation by Coronavirus-19 (COVI-19 or SARS-CoV-2): anti-inflammatory strategies. Conti P, Ronconi G, Caraffa A, Gallenga CE, Ross R, Frydas I, Kritas SK. J Biol Regul Homeost Agents. 2020 March-April,;34(2):327-331.

Mast cells contribute to coronavirus-induced inflammation: new anti-inflammatory strategy. Kritas SK, Ronconi G, Caraffa A, et al. J Biol Regul Homeost Agents. 2020 January-February,;34(1):9-14.

- We appreciate the reviewer pointing important editorials about his work. We, unfortunately, feel the references are editorials and not original articles, only slightly related to the current work, and

we would like to avoid incurring in possible reviewer-pressured citation practices adhering strictly to the BMJ editorial policies.

VERSION 2 – REVIEW

REVIEWER	Besen, Bruno Universidade de Sao Paulo Faculdade de Medicina, Medical Emergencies Discipline
REVIEW RETURNED	13-Jul-2021

GENERAL COMMENTS	The authors have revised the protocol to a large extent to improve its readability and reproducibility. However, I still have a few major concerns regarding this protocol. Now that it's clear this is a prognostic factor (PROGRESS type II) research question (whether inflammatory cytokines are associated with worse outcomes in sepsis or septic shock patients), I am not sure the protocol is completely reproducible in its current form. The rationale of the publication of a systematic review protocol is to enhance transparency and reproducibility of research, making few space for protocol deviations or for the authors to cherry-pick results. My main concerns are as follows: 1) In this being a prognostic research study, if the authors wish to use study-level aggregate measures, since they are dealing with cytokine levels, the authors should better specify how this will be collected and managed in their meta-analysis. As continuous variables? As log-transformed variables? As cut-offs (not very advisable)? How will the authors manage this? The authors should also specify whether they will obtain crude or adjusted effect estimates from individual studies and, if adjusted, adjusted to what illness severity variables. In this case only age and sex probably won't suffice. What adjusted effect measures will be considered adequate? In its current writing, the authors state general statements of meta-analyses, without considering the specificities of their research question. 2) I am still concerned with the proposed subgroup analyses which are based on mean individual level covariates of the studies. How will the authors conduct a subgroup analysis based on mean age < 18 years old or ≥ 18 years old? This doesn't seem feasible for example. Subgroup analyses in meta-analyses aim for exploring heterogeneity. In this case, possible variables would be single center vs. multicenter studies, small sample studies vs. larger sample sizes, high quality vs. low quality studies and so on. Furthermore, this needs to be specified (Example: sample size ≥ 100 vs. < 100) with appropriate explanation and, if later on this is changed, the authors could explain why changes were made with its rationale. Minor issues: - I suggest the authors consider revising the verbal tenses of their manuscript to the future, avoiding past tenses when they are not

	appropriate (Ex: Page 7, line 236 - "quality of included studies was assessed ..."). Positive Issues - The authors specify their research question now delimitated to prognostic factor research - The authors specify a reproducible search strategy - The authors specify a quality assessment tool appropriate for the research question - The authors will present summary of findings table as per GRADE's recommendations Thank you for the opportunity to review your manuscript.

VERSION 2 – AUTHOR RESPONSE

Replies to reviewers' comments:

Comments to the Author:

The authors have revised the protocol to a large extent to improve its readability and reproducibility.

However, I still have a few major concerns regarding this protocol.

Now that it's clear this is a prognostic factor (PROGRESS type II) research question (whether inflammatory cytokines are associated with worse outcomes in sepsis or septic shock patients), I am not sure the protocol is completely reproducible in its current form. The rationale of the publication of a systematic review protocol is to enhance transparency and reproducibility of research, making few space for protocol deviations or for the authors to cherry-pick results.

My main concerns are as follows:

1) In this being a prognostic research study, if the authors wish to use study-level aggregate measures, since they are dealing with cytokine levels, the authors should better specify how this will be collected and managed in their meta-analysis. As continuous variables? As log-transformed variables? As cut-offs (not very advisable)? How will the authors manage this? The authors should also specify whether they will obtain crude or adjusted effect estimates from individual studies and, if adjusted, adjusted to what illness severity variables. In this case only age and sex probably won't suffice. What adjusted effect measures will be considered adequate? In its current writing, the authors state general statements of meta-analyses, without considering the specificities of their research question.

- Page 6, Line 175. We added more specificity on how the cytokine levels will be extracted and analyzed: "We will extract the crude unadjusted mean and standard deviation from each cytokine in all studies. We will then calculate the standard error and log-transformed the values. We will enter the log-transformed mean and standard errors in a random-effect meta-analysis to combine the estimated log pooled values, which will be back-transformed for interpretation."

2) I am still concerned with the proposed subgroup analyses which are based on mean individual level covariates of the studies. How will the authors conduct a subgroup analysis based on mean age < 18 years old or ≥ 18 years old? This doesn't seem feasible for example. Subgroup analyses in

meta-analyses aim for exploring heterogeneity. In this case, possible variables would be single center vs. multicenter studies, small sample studies vs. larger sample sizes, high quality vs. low quality studies and so on. Furthermore, this needs to be specified (Example: sample size ≥ 100 vs. < 100) with appropriate explanation and, if later on this is changed, the authors could explain why changes were made with its rationale.

- Page 8, line 262. We extensively review each potential subgroup analysis based on a pre-specified criteria and clearly explain the type of subgroup analysis that will be conducted. Also, we added a sentence for clarification for continuous variables: “Continuous variables extracted means will be checked for linearity using a two-way scatter plot against log-transformed cytokine mean levels and a coefficient correlation using meta-regression”.

Minor issues:

- I suggest the authors consider revising the verbal tenses of their manuscript to the future, avoiding past tenses when they are not appropriate (Ex: Page 7, line 236 - "quality of included studies was assessed ...").

- Noted. Sentence was corrected as suggested.

Positive Issues

- The authors specify their research question now delimitated to prognostic factor research

- The authors specify a reproducible search strategy

- The authors specify a quality assessment tool appropriate for the research question

- The authors will present summary of findings table as per GRADE's recommendations

- Thank you very much for the kind feedback

Thank you for the opportunity to review your manuscript.

VERSION 3 – REVIEW

REVIEWER	Besen, Bruno Universidade de Sao Paulo Faculdade de Medicina, Medical Emergencies Discipline
REVIEW RETURNED	16-Jul-2021

GENERAL COMMENTS	The authors have addressed all issues.